Non-Profit Reactive Triage Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement and Objective

1.1. Problem Statement

Non-profit organizations receive a large number of support requests in unstructured text, making it difficult to manually classify, prioritize, and respond to them efficiently. This often leads to delays and mismanagement of urgent cases. There is a need for an automated system that can analyze incoming requests, determine their intent and urgency, extract key information, and generate appropriate responses to improve efficiency and ensure timely assistance.

1.2. Project Objective

The objective of this project is to develop an AI-based Non-Profit Support Triage Agent that automates the handling and analysis of support requests received by non-profit organizations. The system processes user messages, classifies them based on intent, determines urgency, extracts key information, and generates appropriate responses to improve efficiency and decision-making.

Key objectives include:

- Automate the classification of user requests into predefined categories
- Identify and prioritize requests based on urgency (high, medium, low)
- Extract important information such as names, dates, and other entities
- Generate context-aware responses using AI
- Reduce manual effort and improve response time
- Provide a scalable and efficient system for handling large volumes of requests

1.3. Scope of the project

The scope of this project is to design and develop an AI-based Non-Profit Support Triage Agent that automates the handling of support requests using natural language processing. The system analyzes text-based inputs such as queries, complaints, and emails, classifies them based on intent and urgency, extracts key information using Named Entity Recognition (NER), and generates context-aware draft responses.

The project is implemented as a prototype within a controlled environment and does not include real-time integration with external databases or emergency services. It focuses on demonstrating the effectiveness of AI-driven automation in improving response efficiency and reducing manual effort.

The system follows a modular and scalable architecture, enabling future enhancements such as integration with real-world non-profit systems, advanced analytics, and improved decision-making capabilities.

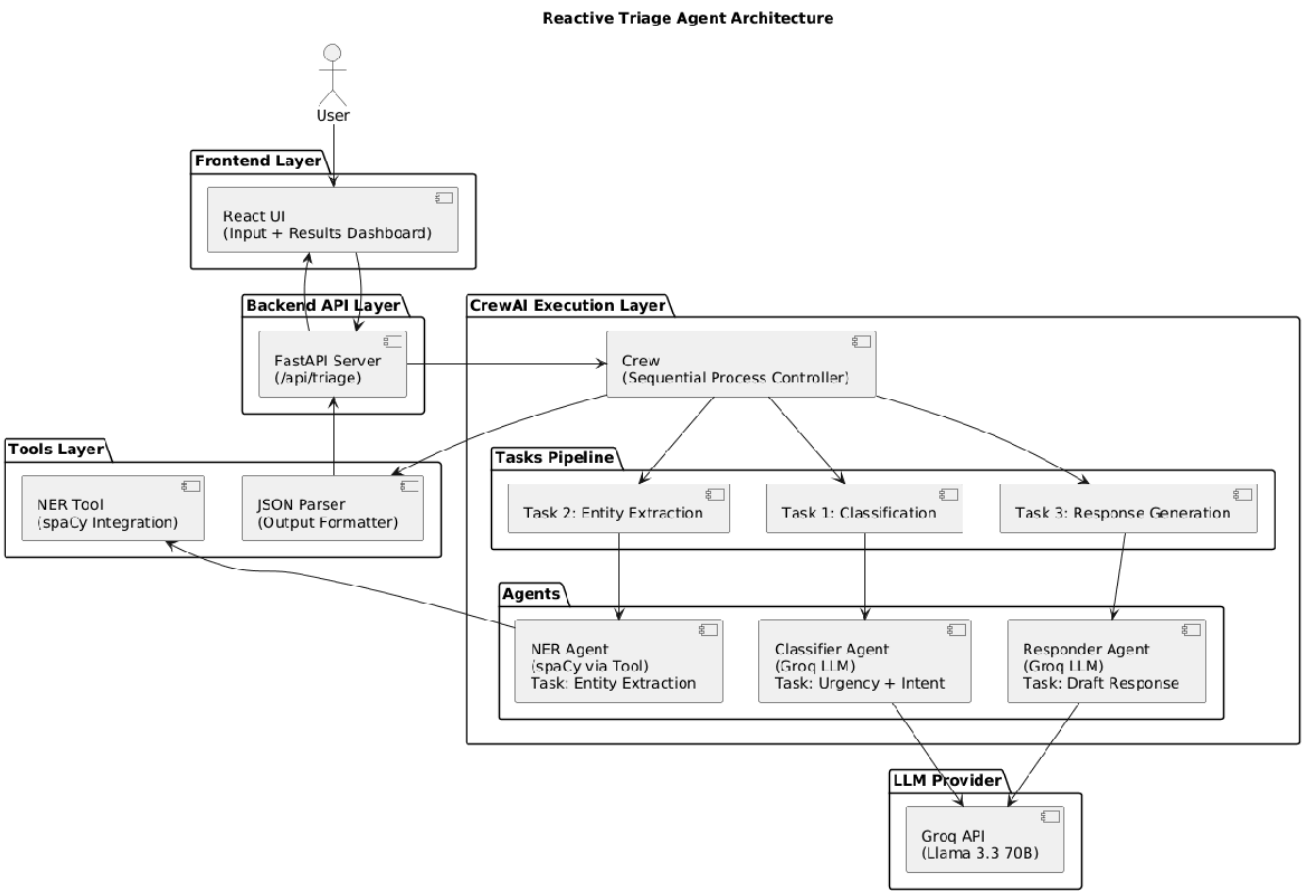
2. Proposed Solution

2.1. Key Features

- **Automated Request Processing:** Handles incoming user messages automatically without manual intervention
- **Intent Classification:** Classifies requests into categories such as food, medical aid, shelter, and education
- **Urgency Detection:** Determines the priority level of requests (high, medium, low)
- **Entity Extraction (NER):** Extracts important details like names, dates, and IDs using NLP techniques
- **AI-Based Response Generation:** Generates context-aware and meaningful replies

- **Multi-Agent Architecture:** Uses multiple agents (classifier, extractor, responder) for structured processing
- **Fast API Backend:** Efficient request handling and routing using FastAPI
- **User-Friendly Interface:** Interactive frontend built with React for easy user interaction
- **Scalable Design:** Modular architecture allows easy expansion and future enhancements
- **Stateless Processing:** Each request is handled independently, improving performance and reliability

2.2. Overall Architecture

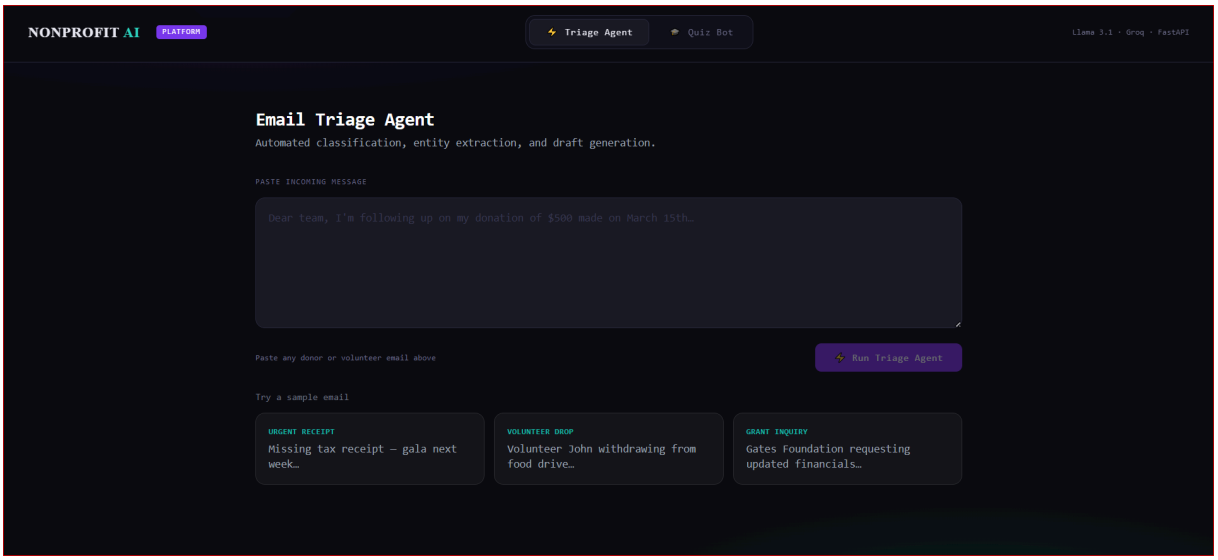


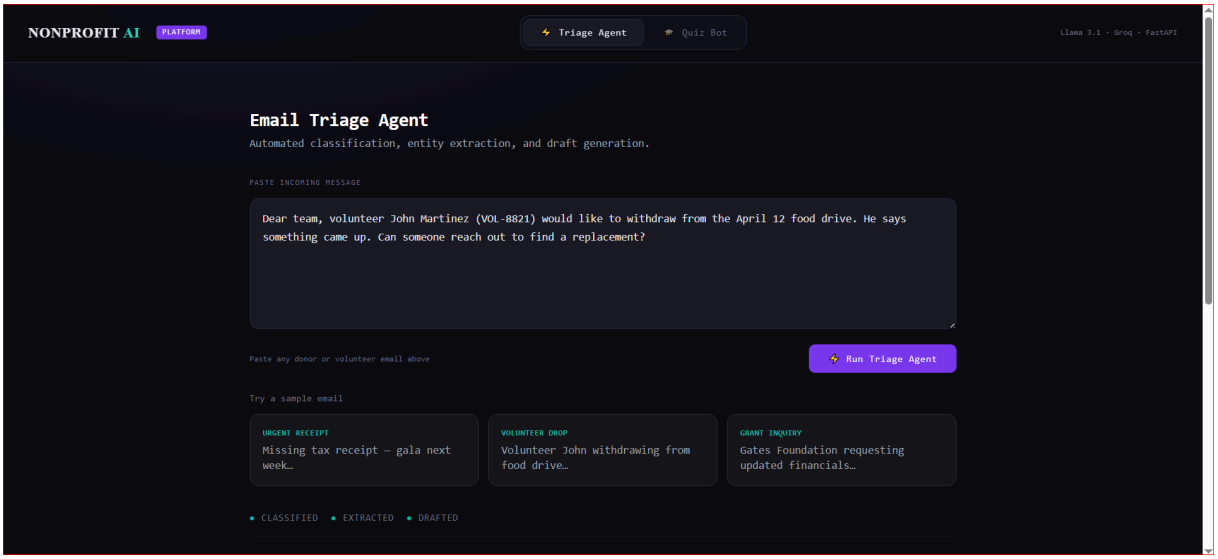
2.3. Tools & Technologies Used

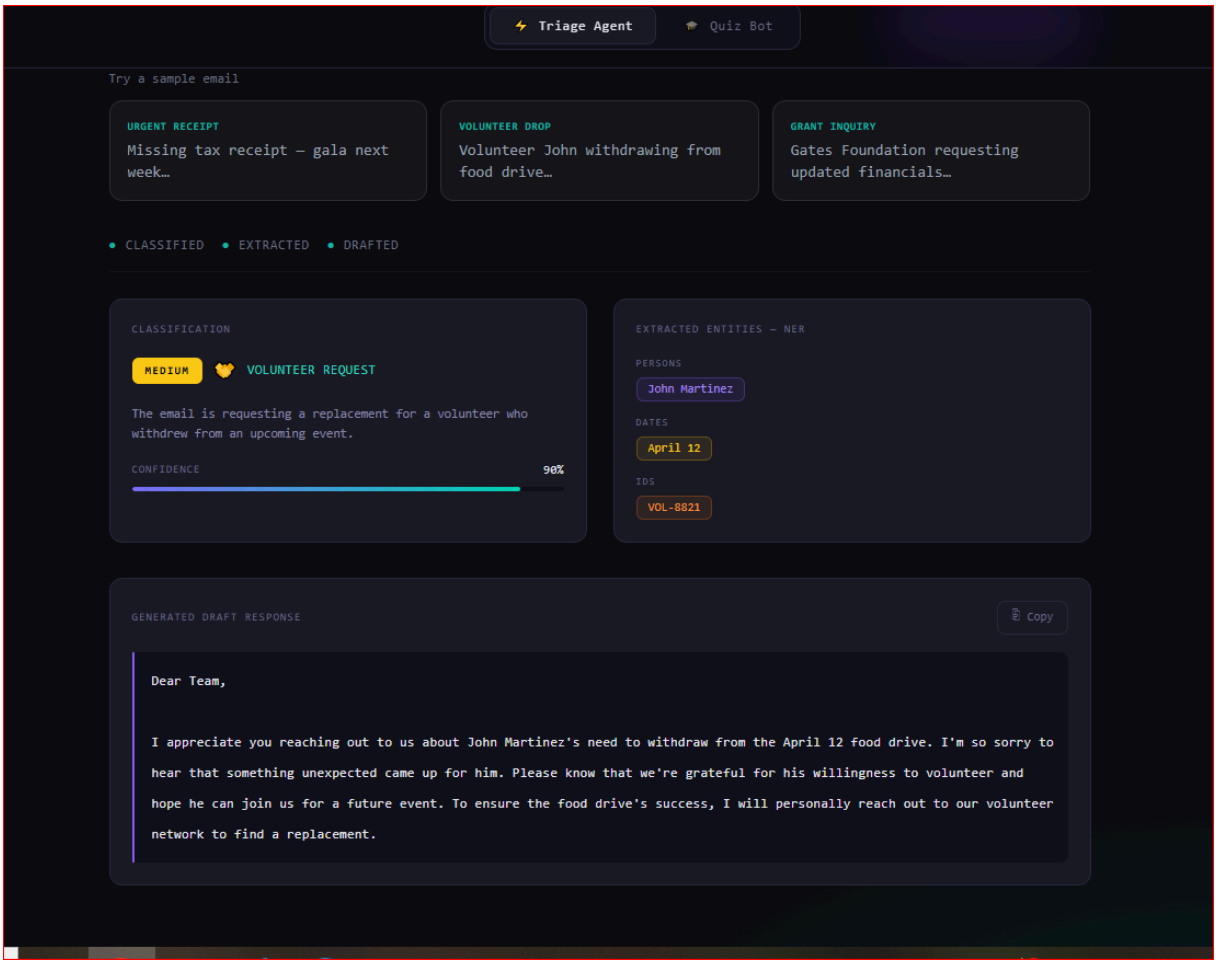
- React – Used for building the user-friendly frontend interface
- FastAPI – Used for developing backend APIs and handling requests
- CrewAI – Used to manage and coordinate multiple AI agents
- LangChain – Used for integrating and managing LLM workflows
- spaCy – Used for Named Entity Recognition (NER)
- Groq – Used for fast LLM processing and response generation
- ChromaDB – Used for storing and retrieving embeddings (optional)
- Uvicorn – Used to run the FastAPI application
- Python – Core language for backend and AI development

3. Results & Output

3.1. Screenshots




NONPROFIT AI
PLATFORM
Triage Agent
Quiz Bot
Llama 3.1 · Groq · FastAPI
Email Triage Agent
Automated classification, entity extraction, and draft generation.
PASTE INCOMING MESSAGE
Dear team, volunteer John Martinez (VOL-8821) would like to withdraw from the April 12 food drive. He says something came up. Can someone reach out to find a replacement?
Paste any donor or volunteer email above
Run Triage Agent
Try a sample email
URGENT RECEIPT
Missing tax receipt – gala next week…
VOLUNTEER DROP
Volunteer John withdrawing from food drive…
GRANT INQUIRY
Gates Foundation requesting updated financials…
CLASSIFIED
EXTRACTED
DRAFTED


Triage Agent
Quiz Bot
Try a sample email
URGENT RECEIPT
Missing tax receipt – gala next week…
VOLUNTEER DROP
Volunteer John withdrawing from food drive…
GRANT INQUIRY
Gates Foundation requesting updated financials…
CLASSIFIED
EXTRACTED
DRAFTED
CLASSIFICATION
MEDIUM
VOLUNTEER REQUEST
The email is requesting a replacement for a volunteer who withdrew from an upcoming event.
CONFIDENCE
90%
EXTRACTED ENTITIES – NER
PERSONS
John Martinez
DATES
April 12
IDS
VOL-8821
GENERATED DRAFT RESPONSE
Copy
Dear Team,
I appreciate you reaching out to us about John Martinez's need to withdraw from the April 12 food drive. I'm so sorry to hear that something unexpected came up for him. Please know that we're grateful for his willingness to volunteer and hope he can join us for a future event. To ensure the food drive's success, I will personally reach out to our volunteer network to find a replacement.

3.2 Key Outcomes

- Successful development of an AI-based system for automating support request handling
- Efficient classification of user queries based on intent and urgency
- Accurate extraction of important information using NLP techniques
- Generation of relevant and context-aware responses using AI
- Reduction in manual effort and faster processing of requests
- Improved prioritization of urgent cases for timely assistance
- Scalable and modular system design for future enhancements
- Enhanced overall efficiency in managing large volumes of support requests

4. Conclusion

The Non-Profit Support Triage Agent successfully demonstrates the application of AI in automating and improving the handling of support requests for non-profit organizations. By combining classification, entity extraction, and response generation, the system is able to process user queries efficiently and reduce the need for manual intervention.

The project enhances the ability to prioritize urgent cases, ensures faster response times, and improves overall workflow management. Its modular and scalable design makes it flexible for future enhancements such as real-time integration, multilingual support, and advanced analytics. Although currently a prototype, the system provides a strong foundation for developing intelligent and efficient support systems in real-world non-profit environments.

5. Future Scope & Enhancements

The current system provides a strong foundation for automating support request handling; however, it can be further improved and expanded with advanced features to enhance its real-world applicability.

Future Scope:

- Integration with real-time non-profit databases and emergency services
- Support for multiple languages to handle diverse users
- Addition of voice input and speech-to-text capabilities
- Development of mobile applications for wider accessibility
- Implementation of real-time notifications and alerts for urgent cases

Enhancements:

- Adding authentication and user management for better security
- Improving AI accuracy with advanced models and training data
- Integration with external APIs for resource allocation and tracking
- Enhanced analytics and dashboard for monitoring requests
- Incorporation of feedback mechanisms to improve system performance